

CS 576 – Assignment 1

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Assigned on Mon Jan 26, 2025,

Solutions due on Mon Feb 16, 2025 - by 5:00 pm afternoon

Late policies - None

Theory Part

Question 1

Suppose a camera has 450 lines per frame, 520 pixels per line, and 25 Hz frame rate. The color sub sampling scheme is 4:2:0, and the pixel aspect ratio is 16:9. The camera uses interlaced scanning, and each sample of Y, Cr, Cb is quantized with 8 bits

- What is the bit-rate produced by the camera?
- Suppose we want to store the video signal on a hard disk, and, to save space, the signal is re-quantized so that each channel (Y, Cr, Cb) uses 6 bits. What is the minimum size of the hard disk required to store 10 minutes of video?

2. bits/frame = $\checkmark \times 6$ bits/10 mins = $\checkmark \times 600$

Question 2

The following sequence of real numbers has been obtained sampling an audio signal: 1.8, 2.2, 2.2, 3.2, 3.3, 3.3, 2.5, 2.8, 2.8, 2.8, 1.5, 1.0, 1.2, 1.2, 1.8, 2.2, 2.2, 2.2, 1.9, 2.3, 1.2, 0.2, -1.2, -1.2, -1.7, -1.1, -2.2, -1.5, -1.5, -0.7, 0.1, 0.9. Quantize this sequence by dividing the interval $[-4, 4]$ into 32 uniformly distributed levels (place the level 0 at -3.75, the level 1 at -3.5, and so on. This should simplify your calculations).

- Write down the quantized sequence.
- How many bits do you need to transmit it?

1. $g(k) = -3.75 + 0.25 \times k$
 $k = \left\lfloor \frac{x+3.75}{0.25} \right\rfloor$

value = $-3.75 + 0.25 \times k$

2. bit required
 $= \log_2 32 = 5$
 $32 \times 5 = 160$
 $\uparrow$
 sample count

Question 3

Temporal aliasing can be observed when you attempt to record a rotating wheel with a video camera. In this problem, you will analyze such effects. Assume there is a car moving at 36 km/hr and you record the car using a film, which traditionally record at 24 frames per second. The tires have a diameter of 0.4244 meters. Each tire has a white mark to gauge the speed of rotation. Assume that the tire rotates without skidding.

- If you are watching this projected movie in a theatre, what do you perceive the rate of tire rotation to be in rotations/sec?
- If you use your camcorder to record the movie in the theater and your camcorder is recording at one third film rate (ie 8 fps), at what rate (rotations/sec) does the tire rotate in your video recording
- If you use an NTSC camera with 30 fps, what is the maximum speed that the car can go at so that you see no aliasing in the recording.

1. $\frac{36000}{3600} = 10 \text{ m/sec}$

$\frac{10}{0.4244\pi} = \frac{\text{rotations}}{\text{sec}} = 7.5 \text{ rot/sec}$

Nyquist limit = $\frac{24}{2} = 12$

$12 > 7.5$

So we perceive 7.5 rot/sec, no aliasing

2. Aliasing occurs since $7.5 \times 2 > 8$

Ans: $7.5 - 8 = -0.5 \text{ rot/sec}$

3. We can have 15 rot/sec .

$$0.4244\pi \times 15 = 72 \text{ km/hr}$$